



Department for
Science, Innovation
& Technology

Aligning an Amazon Web Services (AWS) data migration strategy with Secure by Design

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Background and purpose

What AWS data migration involves

Data migration to Amazon Web Services (AWS) is the process of moving data, applications and workloads from on-premises systems or other cloud environments into AWS infrastructure.

This typically includes:

- assessing existing data sources
- choosing an appropriate migration approach
- using cloud-native tools to transfer data securely and efficiently

The aim is to maintain data integrity, minimise downtime and support business continuity.

Purpose of this document

This document supports government organisations in aligning their AWS data migration activities with the [Secure by Design](#) approach.

Secure by Design is a government framework that embeds cyber security into digital services from the outset. By integrating security as a proactive and integral part of migration planning and delivery, organisations can reduce risk and strengthen resilience throughout the transition to cloud services.

This guidance provides a structured approach for selecting suitable migration methods, tools and timelines based on technical and operational requirements.

AWS migration services

AWS offers a range of migration services to support secure and low-disruption migrations, including:

- [AWS Database Migration Service \(AWS DMS\)](#) for database migration with minimal downtime
- [AWS Schema Conversion Tool \(AWS SCT\)](#) for converting database schemas to AWS-compatible formats
- [AWS Snowball](#) and [AWS DataSync](#) for large-scale data transfers
- [AWS Transfer Family](#) for file-based migrations using SFTP, FTPS, and FTP protocols

Secure by Design on security.gov.uk

- [Secure by Design policy](#)
- [Implementing Secure by Design](#)
- [Guide to adopting Secure by Design](#)

1. Migration strategy framework

The migration strategy you choose should align with Secure by Design principle 6, [Design flexible architectures](#).

The commonly-used '6 Rs' model provides a structured way to classify migration approaches:

- **rehosting** may move workloads quickly, but can also carry forward existing vulnerabilities and technical debt
- **replatforming** introduces some optimisation, often improving manageability without a full redesign
- **refactoring** enables deeper transformation, allowing security controls to be embedded natively within application and data layers
- **repurchasing** replaces existing applications with Software as a Service (SaaS) products or newer solutions, often improving scalability, security and ease of management
- **retiring** removes systems that are no longer needed
- **retaining** keeps stable legacy environments where migration risk outweighs the benefit

Architectural flexibility should be viewed as a core part of long-term risk management. Systems should be designed to accommodate changes in:

- security requirements
- compliance
- business needs

A flexible architecture makes it easier to respond effectively without major redesign or service disruption.

2. Migration planning

Migration planning should embed security considerations from the outset and maintain them throughout delivery. This section outlines key activities aligned with [Secure by Design principles](#).

2.1 Discovery and assessment

Discovery activities should align with the Secure by Design principle 3, [Adopt a risk-driven approach](#).

Architects need a detailed understanding of:

- data assets

- application dependencies
- infrastructure relationships
- operational constraints

For more about this, refer to the Secure by Design activity [Documenting service assets](#).

Data classification should be carried out to determine sensitivity and criticality. This should be complemented by structured threat modelling to identify:

- potential attack paths
- misuse scenarios
- risks introduced by migration activities

You should also conduct dependency mapping. This helps make sure changes to one system do not unintentionally disrupt others during migration.

This phase should also consider shadow data and undocumented data sets. These can introduce hidden risks and may not be visible through technical discovery alone. Identifying them often requires strong engagement with business and technical stakeholders. Early identification reduces uncertainty during migration and improves overall security and governance.

For more information about data classification, refer to the [Government Security Classifications Policy](#).

2.2 Define success criteria

Success criteria should reflect your organisation's risk profile, threat landscape and security obligations. They should also support Secure by Design principle 4, [Design usable security controls](#).

Security controls should be effective in protecting systems while remaining practical to operate, monitor and maintain in production environments.

Success criteria should include measurable outcomes such as:

- secure authentication behaviour
- encryption enforcement
- audit logging completeness
- monitoring coverage

These criteria should be built into both functional and non-functional testing frameworks. This should help make security validation part of migration success rather than a separate activity.

2.3 Migration waves

Use a phased migration approach so that you can test and validate changes in stages. This aligns with Secure by Design principle 9, [Embed continuous assurance](#).

Start with low-risk or non-critical systems. This allows you to:

- validate migration tooling
- establish baseline performance
- test security controls, operational processes and system behaviour in real conditions

Later phases should include more complex workloads, incorporating lessons learned from earlier phases.

Treat each wave as a controlled validation step rather than just a delivery milestone.

3. AWS migration tools overview

How you select and configure AWS migration tools should align with Secure by Design principle 2, [Sourcing secure technology products](#).

When choosing tools, do not only consider functional requirements. You should also think about:

- security maturity
- how well the tool integrates with existing platforms
- visibility into its activity and behaviour

You should configure services like [AWS Database Migration Service](#) and [AWS DataSync](#) to work with your existing security controls. This includes:

- identity management systems
- encryption key management
- logging and monitoring platforms

You should review and adjust default configurations so they're aligned with organisational security baselines and regulatory requirements.

Tooling strategies should also cover life cycle management. This includes:

- patching
- version upgrades
- decommissioning migration tools when they're no longer needed

This prevents tooling from becoming a long-term unmanaged risk within the environment.

4. Network considerations

Design secure and controlled connectivity

Network architecture should align with Secure by Design principles 7, [Minimise the attack surface](#), and 8, [Defend in depth](#).

Connectivity between on-premises and AWS environments should be designed to support secure, controlled and observable data flows.

Segment and monitor network traffic

Migration traffic should be isolated from production workloads wherever possible.

Apply traffic inspection, logging and anomaly detection to identify unusual patterns or potential data exfiltration attempts.

At the same time, reduce unnecessary network exposure by limiting routes, endpoints and externally accessible interfaces.

Build resilience based on risk

Network resilience should be incorporated into design decisions based on risk assessment or threat modelling. This helps determine how connectivity failures would affect migration and ongoing operations.

Rather than applying the same resilience measures everywhere, design them proportionately according to:

- system dependencies
- business criticality
- recovery objectives

Where needed, use redundant connectivity paths and failover mechanisms to reduce the risk of migration disruption caused by infrastructure failure or service degradation.

5. Database migration strategies

Apply layered security controls

Database migration should align with the Secure by Design principle 8, [Defend in depth](#).

Structured data systems require layered protections. This must include:

- securing the storage layer with encryption at rest
- controlling access through strong identity and role-based permissions
- protecting data replication and transfers with encrypted connections and restricted pathways
- enabling monitoring and logging to detect unusual activity or potential security breaches

Manage schema transformation carefully

Schema transformation is required because source and target systems often differ in structure, data types and constraints. This means data must be adapted to remain usable and consistent after migration.

This process should align with Secure by Design principle 10, [Make changes securely](#). You should validate changes through testing to avoid introducing vulnerabilities or breaking data integrity.

If you're using continuous replication strategies, monitor them to detect:

- inconsistencies
- latency issues
- unauthorised modifications

For more about this, refer to AWS's guidance on [schema conversion](#).

Balance performance and security

You should also consider database performance tuning as part of migration design, but make sure that this does not weaken security controls or bypass governance processes.

Performance optimisation, configuration review and any additional security hardening should be completed as part of pre-migration readiness.

6. File and object storage migration

Restrict access and reduce exposure

File and object storage migration should align with Secure by Design principle 7, [Minimise the attack surface](#).

Configure storage systems so they can only be accessed by identities and services which need it.

Apply consistent security controls

Storage services should be configured with:

- private access settings
- encryption enforcement
- strict permission boundaries

These should be applied consistently across all storage resources.

Review access regularly

Access policies should be reviewed on a regular basis so they're aligned with current operational requirements. This also prevents privilege creep, where users or services retain access they no longer require.

For more on this, refer to the National Cyber Security Centre (NCSC) guidance on [privileged access management](#).

7. Data validation

Validate data throughout the migration

Data validation processes should align with Secure by Design principle 9, [Embed continuous assurance](#).

Validation should happen throughout the migration life cycle rather than treated as a single step at the end of migration.

This means validating data quality:

- before migration
- during transfer and transformation
- after migration
- during ongoing operation

Use automation where possible

You should use automated validation to continuously reconcile source and target data sets.

This should cover:

- integrity checks
- completeness

- structural consistency

Handling exceptions

Validation processes should include exception handling workflows. When issues are detected, teams should be able to:

- investigate quickly
- apply fixes without delaying the migration progress

8. Cutover strategies

Plan controlled and reversible transitions

Cutover strategies should align with Secure by Design principle 10, [Make changes securely](#).

The move from the source to target environments should be carefully controlled, validated and reversible where possible.

Include validation and rollback mechanisms

Cutover design should include:

- rollback strategies
- validation gates
- clear operational communication plans

These controls help reduce the impact of any unexpected behaviour and support faster recovery.

Align cutover with business operations

Cutover timing should take account of business needs. Plan transitions so that there's minimal impact to critical services and service continuity.

9. Monitoring and observability

Monitoring and observability should align with Secure by Design principle 5, [Build in detect and respond security](#).

Systems should be designed to provide real-time visibility into:

- operational behaviour
- security events
- performance metrics

Logging, monitoring and alerting systems should be integrated across all migration components so that coverage is consistent. Logs should be protected against tampering and retained in accordance with governance requirements.

Observability design should also support forensic investigation, not just day-to-day monitoring. This means events can be reconstructed if there's a failure or security incident.

10. Automation and infrastructure as code

Automation and infrastructure as code (IaC) should align with Secure by Design principle 10, [Make changes securely](#).

By using automated deployment and configuration processes you can reduce human error and improve consistency across environments.

Creating traceability

Infrastructure and security controls defined as code should be version-controlled, reviewed and tested before deployment.

This provides traceability for changes and helps you consistently enforce security standards across environments.

Validate security within deployment pipelines

Automation pipelines should include security validation stages to detect misconfigurations or policy violations before changes are applied.

11. Testing strategy

Test throughout the migration life cycle

Testing strategies should align with Secure by Design principle 9, [Embed continuous assurance](#).

Testing should start early in the migration life cycle to flag up both issues and solutions. This makes them easier to fix and avoids relying on pre-production validation.

Use different types of testing

Integrate functional, performance and security testing into migration workflows. Vulnerability scanning and penetration testing should be used to identify weaknesses before production cutover.

Use representative test environments

Test environments should closely match production configurations so that results are representative and reliable.

12. Migration execution

Data migration execution should be guided by a risk-based assessment and must align with Secure by Design principle 5, [Build in detect and respond security](#).

Execution activities should be continuously monitored to identify and address issues in real time. Monitoring, alerting and incident response should be proportionate to:

- the sensitivity of the data
- the criticality of the systems involved
- the level of threat exposure

Operational teams should maintain visibility over data transfer, validation processes and system behaviour. You should establish clear escalation and communication channels to ensure a rapid response to incidents.

Proper governance should be in place so that any deviations from planned behaviour are assessed promptly and managed through agreed control processes.

13. Post-migration activities

You should review and refine systems to make sure they can evolve as requirements and threats change.

Once migration is complete, carry out:

- performance optimisation
- security hardening
- configuration reviews

Capture lessons learned and apply them to future migration waves or transformation initiatives. For more on this, refer to GOV.UK's [Lessons Management Best Practice Guidance](#).

Manage legacy system decommissioning carefully so that removal of residual data, credentials and connectivity paths is secure. For more on this, refer to the GOV.UK guidance on [managing legacy technology](#).

More information

- [Secure by Design principles \(security.gov.uk\)](#)
- [Secure by Design activities \(security.gov.uk\)](#)
- [Threat modelling \(NCSC\)](#)
- [Cloud security guidance \(NCSC\)](#)
- [Managing observability \(security.gov.uk\)](#)
- [Encryption and data protection \(ICO\)](#)